Barren plateaus in parametrized quantum circuits

The **barren plateaus** (*BP*) phenomenon in quantum machine learning is a mathematically grounded problem appearing in the training of parametrized quantum circuits (*PQC*). Mathematically, it stems from the high-dimensional nature of quantum states and the random initialization of circuits, leading to exponentially small gradients. It was initially introduced in [1], see [2] for a more recent review.

This contrasts with *classical machine learning*, where vanishing gradients are more architecturally driven and can be mitigated through established techniques (skip connections, better random initialization, etc).

Understanding and addressing barren plateaus is an essential objective for near-term quantum machine learning.

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> H-DES highlights barren plateaus as a serious obstacle to convergence, reinforcing the need for strategies to overcome this issue in practical algorithms.

The loss landscape of parametrized quantum circuits

The BP phenomenon states that the gradients needed to adjust the parameters of quantum circuits become very small. This makes it tough to train these circuits, as the loss function becomes flat.

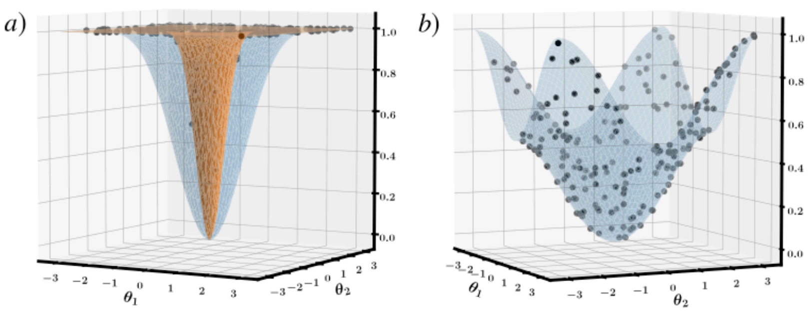

Fig. 2 Cost function landscapes. a Two-dimensional cross-section through the landscape of $C_G = 1 - \prod_{j=1}^{n} \cos^2(\theta^j/2)$ for $n = 4$ (blue) and $n = 24$ (orange). **b** The same cross-section through the landscape of $C_L = 1 - \frac{1}{n}\sum_{j=1}^{n} \cos^2(\theta^j/2)$ is independent of n. In both cases, 200 Haar distributed points are shown, with very few (most) of these points lying in the valley containing the global minimum of C_G (C_L).

Consider a PQC with output state $|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|\psi_0\rangle$, where $U(\boldsymbol{\theta})$ is the unitary implemented by the circuit and $|\psi_0\rangle$ is a fixed initial state (e.g. $|0\cdots0\rangle$). We define a **cost function** $C(\boldsymbol{\theta})$ as the expectation value of some Hermitian observable O on the output state:

$$C(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|O|\psi(\boldsymbol{\theta})\rangle = \mathrm{Tr}\left[O\rho(\boldsymbol{\theta})\right]$$

where $\rho(\boldsymbol{\theta}) = |\psi(\boldsymbol{\theta})\rangle\langle\psi(\boldsymbol{\theta})|$ is the density matrix of the output state. This formulation covers many variational algorithms; for example, in a variational quantum eigensolver O could be a Hamiltonian whose ground state we seek, and in a quantum classifier O could be an observable encoding the training loss.

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> In H-DES, the observable O encodes the error in satisfying the differential equation, and the cost is the expectation value of this error operator. The training process minimizes this cost to solve the PDE, and thus is likewise vulnerable to barren plateaus.

The cost $C(\boldsymbol{\theta})$ is a smooth function of the parameters, and our interest is in its *gradient* $\nabla_{\boldsymbol{\theta}} C$, which guides how we update parameters in gradient-based optimization.

For a given parameter θ_i, the partial derivative of the cost is obtained by differentiating under the expectation value:

$$\frac{\partial C}{\partial \theta_i} = \frac{\partial}{\partial \theta_i} \langle \psi(\boldsymbol{\theta}) | \, O \, | \psi(\boldsymbol{\theta}) \rangle.$$

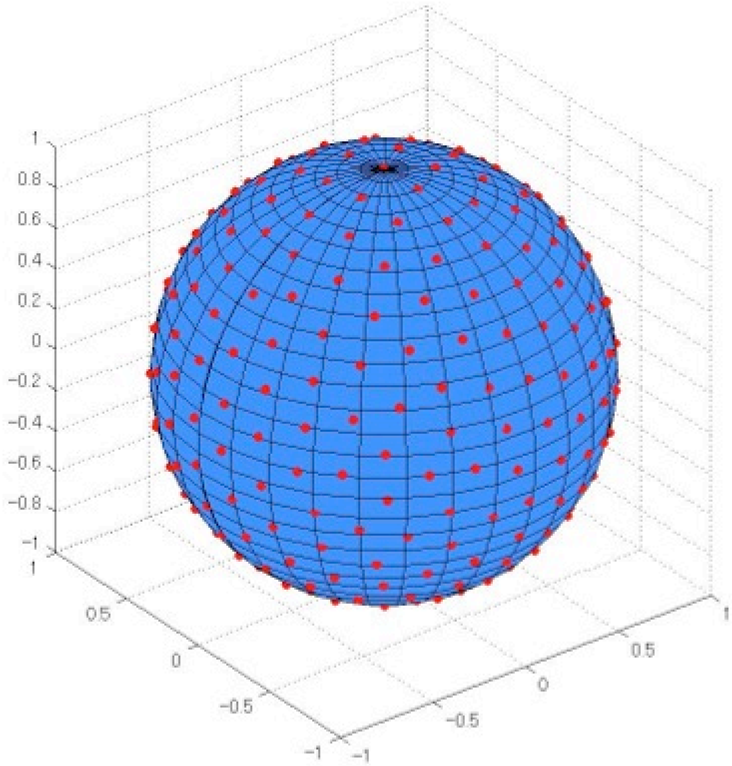
Using the chain rule for unitary dependence on θ_i, one can show this derivative can be written in terms of the *generator* of the parameterized gate. Typically, if θ_i enters the circuit via a unitary of the form $U_i(\theta_i) = e^{-i\theta_i V_i}$ (where V_i is some Hermitian operator, e.g. a *Pauli matrix* for rotation gates), then:

$$\frac{\partial C}{\partial \theta_i} = \langle \psi(\boldsymbol{\theta}) | U^\dagger(\boldsymbol{\theta}) [V_i, O] U(\boldsymbol{\theta}) | \psi_0 \rangle.$$

Now, let us consider *random quantum circuits* and their implications for gradient behavior. By a "random circuit," we mean that the parameters (and possibly even the circuit structure) are initialized at random, such that the unitary $U(\boldsymbol{\theta})$ is drawn from some distribution over the unitary group. In the extreme case of a *Haar-random* circuit, U is distributed according to the uniform (Haar) measure on the unitary group – intuitively, the output state $|\psi(\boldsymbol{\theta})\rangle$ is a random pure state uniformly distributed on the Bloch sphere of 2^n dimensions (for n qubits). More practically, even if the circuit has a specific fixed architecture, a *sufficiently deep and entangling* random circuit often behaves like a random unitary on its n qubit Hilbert space. Such circuits are said to form a *unitary 2-design* when their statistics up to second moments match those of the Haar ensemble.

Spherical and unitary 2-designs

A *spherical t-design* is a finite subset of points of the unit sphere of $\mathbb{C}^d$ such that integration with respect to a given finite measure on the design (usually the uniform meaure) of a polynomial of *degree at most t* yields the same result as integration with respect to the Haar measure (i.e. uniform measure) of $\mathbb{C}^d$. Below is an example of a spherical 16-design for the unit sphere of $\mathbb{R}^3$ having 289 points, see https://www.polyu.edu.hk/ama/staff/xjchen/sphdesigns.html



A *unitary t-design* is a finite subset of the unitary group $\mathcal{U}_d$ having a similar property: integrating any polynomial function of degree at most t in the entries of a unitary matrix (and its complex conjugates) over a measure on the design produces the same result as the integral with respect to the Haar measure on $\mathcal{U}_d$.

Random quantum circuits on n qubits in a *brickwork architecture* of depth $O(nt^{5+o(1)})$ are approximate unitary t-designs [3]. The proof goes through lower bounds on the *spectral gap of moment operators* for local random quantum circuits.

Gradients concentrate around zero for random circuits

Let us first examine the expected value of the gradient of a cost function when the circuit parameters are randomized. Suppose $U(\boldsymbol{\theta})$ is drawn from an ensemble that is *sufficiently random* (specifically, forms at least a 2-design). One can show that the *average gradient is zero*:

$$\mathbb{E}_{\boldsymbol{\theta}}\left[\frac{\partial C}{\partial \theta_i}\right] = 0$$

for any parameter θ_i. This is intuitively because for a random symmetric distribution of unitaries, there is no preferred direction in parameter space – the cost is as likely to increase as to decrease with a small perturbation, yielding zero average slope. A more rigorous argument uses the Haar integration: under the Haar measure, the distribution of U is unitarily invariant, and one can show

$$\mathbb{E}[U^\dagger O U] = (\mathrm{Tr} O/2^n) I.$$

If O is traceless (which we can assume w.l.o.g. by shifting the cost by a constant if needed), then $\mathbb{E}[U^\dagger O U] = 0$. This implies $\mathbb{E}[\partial C/\partial\theta_i] = 0$ as it is an expectation value of some traceless operator in a random state.

More importantly, not only is the mean gradient zero, but typical fluctuations are extremely suppressed. In high-dimensional quantum state space, *concentration of measure* occurs, meaning that most states yield nearly the same expectation value for any given observable. Formally, the probability that a random state's observable expectation deviates from the mean by more than a small ε is exponentially small in the number of qubits . This is a consequence of Lévy's lemma (a result from measure concentration), which in our context implies that $\langle\psi(\boldsymbol{\theta})|O|\psi(\boldsymbol{\theta})\rangle$ is highly concentrated around $\mathrm{Tr}(O)/2^n$.

Variance of gradient decays exponentially with number of qubits

To make the above statements quantitative, we now calculate (or at least sketch the derivation of) the **variance** of the gradient for random circuits. Since the mean is zero, the variance is just the expected value of the squared gradient:

$$\mathrm{Var}\Big(\frac{\partial C}{\partial\theta_i}\Big) = \mathbb{E}\Big[\Big(\frac{\partial C}{\partial\theta_i}\Big)^2\Big].$$

For a *Haar-random state* $|\varphi\rangle$ on n qubits (i.e. a randomly distributed vector on the unit sphere of $\mathbb{C}^{2^n}$), the expectation $\langle\varphi|O|\varphi\rangle$ has a well-known variance. Assuming O is normalized appropriately (e.g., $|O| = 1$ or O is a sum of Paulis), one finds, using the Weingarten calculus:

$$\mathrm{Var}_{|\varphi\rangle\sim\mathrm{Haar}}\big(\langle\varphi|O|\varphi\rangle\big) = \frac{\mathrm{Tr}(O^2) - (\mathrm{Tr}O)^2/2^n}{2^n(2^n+1)}.$$

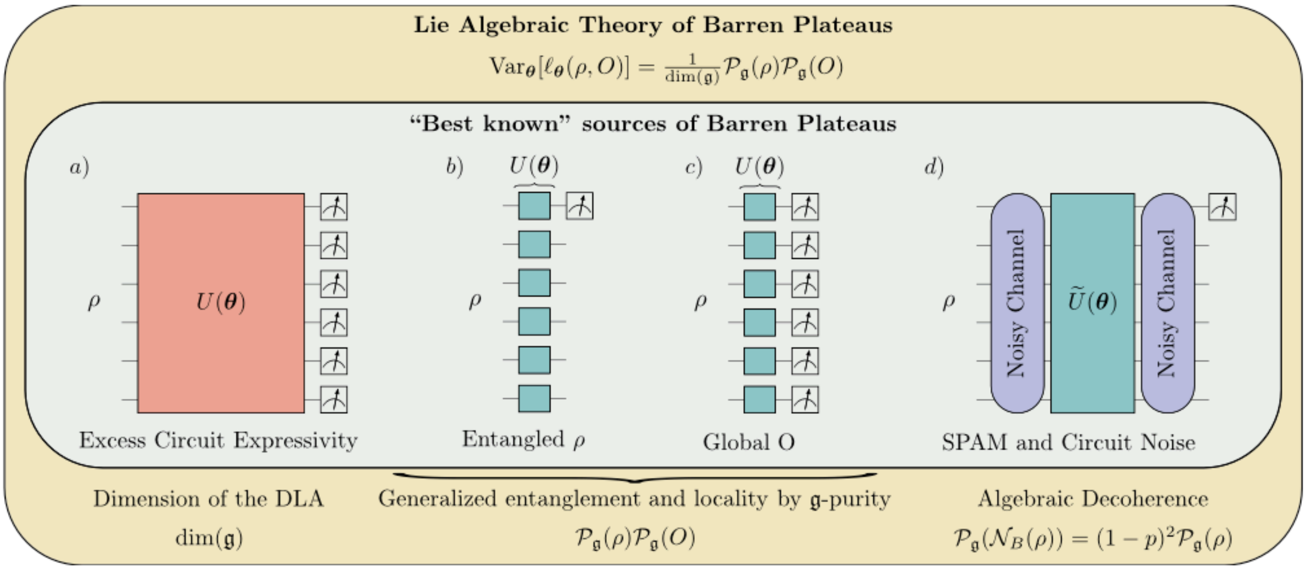
If O is traceless (so that the mean is zero), this simplifies to $\mathrm{Var}(\langle O\rangle) = \frac{\mathrm{Tr}(O^2)}{2^n(2^n+1)}$. In many cases O might be a Pauli observable or a tensor product of Paulis, which are symmetries (i.e. ±1 observables) and satisfy $\mathrm{Tr}(O^2) = 2^n$. In that case, $\mathrm{Var}(\langle O\rangle) = \frac{2^n}{2^n(2^n+1)} \approx \frac{1}{2^n}$ for large n. This indicates that a random state's expectation value of a traceless observable is typically of order $2^{-n/2}$.

In summary, for random or highly expressive PQCs, we find a *barren plateau*: almost all directions in parameter space have essentially zero gradient. The expected gradient is zero, and the probability of seeing any appreciable gradient falls off exponentially with n. This exponential concentration of cost values (and thus flatness of the landscape) is the essence of the barren plateaus phenomenon.

A Lie algebraic approach to BP

In [4] the authors propose a novel, rather general, mathematical approach to Barren Plateaus: the study of the **dynamical Lie algebra** (DLA). It unifies several sources of BP previously discussed in the literature, pinpointing the exact mathematical cause of BP. For example, it replaces

- the expressiveness of the PQC (2-design property) with the *dimension of the DLA* $\dim \mathfrak{g}$
- global vs local observables cause of BP with the $\mathfrak{g}$-*purity*
- entanglement of the quantum state ρ with its *generalized entanglement*



Dynamical Lie Algebra (DLA)

Given a parameterized quantum circuit

$$U(\theta) = \prod_{\ell=1}^{L} e^{iH_\ell \theta_\ell},$$

its generators are the Hermitian operators

$$\{H_1, H_2, \ldots\}.$$

The **dynamical Lie algebra** is defined as the Lie closure of these generators:

$$\mathfrak{g} = \mathrm{Lie}\{iH_1, iH_2, \ldots\}.$$

This means that $\mathfrak{g}$ is the smallest subspace of $u(2^n)$ closed under commutators that contains the set

$$\{iH_\ell\}.$$

The authors further show that $\mathfrak{g}$ admits a decomposition into commuting ideals:

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_{k-1} \oplus \mathfrak{g}_k,$$

where for $j = 1, \ldots, k-1$, $\mathfrak{g}_j$ are simple Lie algebras and $\mathfrak{g}_k$ is abelian (the *center*).

Examples from the Paper

- *Highly Expressive Circuit:* When the circuit forms a 2-design over $SU(2^n)$, one has

$$\mathfrak{g} = \mathfrak{su}(2^n)$$

with dimension $\dim(\mathfrak{g}) = 4^n - 1$, implying an exponential expressiveness.
- *Local Single-Qubit Gates:* For circuits composed solely of single-qubit gates, the DLA takes the form

$$\mathfrak{g} = \bigoplus_{j=1}^{n} \mathfrak{su}(2),$$

which scales only polynomially in n.

Loss Function Variance in Terms of the DLA

The loss function is defined by

$$\ell_\theta(\rho, O) = \mathrm{Tr}\left[U(\theta)\rho U^\dagger(\theta)O\right],$$

with ρ the initial state and O the measurement observable. The variance is

$$\mathrm{Var}_\theta[\ell_\theta(\rho, O)] = \mathbb{E}_\theta[\ell_\theta(\rho, O)^2] - \left(\mathbb{E}_\theta[\ell_\theta(\rho, O)]\right)^2.$$

Originally, the BP phenomenon was concerned with the variance of the *partial derivatives* of the cost function. Later, it was shown in [5] that this is equivalent to the vanishing of the variance of the *cost function* itself.

A key result (Theorem 1 in the paper) shows that **if either ρ or O lie in the DLA** (i.e. are “aligned” with $\mathfrak{g}$), then for a *simple* DLA one obtains:

$$\boxed{\mathrm{Var}_\theta[\ell_\theta(\rho, O)] = \frac{P_{\mathfrak{g}}(\rho)\, P_{\mathfrak{g}}(O)}{\dim(\mathfrak{g})},}$$

where the $\mathfrak{g}$*-purity* is defined as

$$P_{\mathfrak{g}}(H) = \mathrm{Tr}\left[H_{\mathfrak{g}}^2\right],$$

with $H_{\mathfrak{g}}$ the projection of H onto the complexification of $\mathfrak{g}$.

For a DLA decomposing as

$$\mathfrak{g} = \bigoplus_{j=1}^{k-1} \mathfrak{g}_j \oplus \mathfrak{g}_k,$$

the variance splits as a sum over the non-abelian components:

$$\mathrm{Var}_\theta[\ell_\theta(\rho, O)] = \sum_{j=1}^{k-1} \frac{P_{\mathfrak{g}_j}(\rho)\, P_{\mathfrak{g}_j}(O)}{\dim(\mathfrak{g}_j)}.$$

Implications for Barren Plateaus

A BP occurs when the loss function's variance vanishes exponentially with the system size. Two main factors determine this:

- *Expressiveness of the Circuit:* When

  $$\dim(\mathfrak{g}) \in \Omega(b^n)$$

  for some $b > 1$, the variance is suppressed as

  $$\mathrm{Var}_\theta[\ell_\theta(\rho, O)] \sim \frac{1}{\dim(\mathfrak{g})},$$

  leading to barren plateaus.
- *Generalized Entanglement and Locality:* Low $\mathfrak{g}$-purity, i.e. $P_\mathfrak{g}(\rho)$ or $P_\mathfrak{g}(O)$ being exponentially small (which corresponds to high generalized entanglement or high nonlocality), also forces the variance to vanish.

Thus, the paper **unifies various sources of barren plateaus**: circuits that are too expressive (large DLA), highly entangled initial states, and nonlocal measurement operators all drive the loss variance toward zero, making the optimization landscape flat.

The authors also consider **SPAM noise**: *State Preparation And Measurement noise*. These errors occur during the initialization and final measurement of quantum states.

SPAM noise can be divided into two main types:

State Preparation Errors:
• Occur when initializing a quantum system into a desired state
• Due to imperfect hardware, the prepared state may deviate from the ideal target
• Causes include decoherence, crosstalk, and environmental noise

Measurement Errors:
• Occur when reading out the quantum state at the end of a computation.
• Instead of getting the correct measurement outcome, noise causes misclassification.

Mitigation

There are three primary causes of BPs:

1. Gradient Vanishing due to Deep Circuits
In deep quantum circuits with many qubits, random parameter initialization can make the circuit act like a *random unitary* over a high-dimensional Hilbert space (*2-design property*).

Mitigation techniques:

- Shallow or Layered Circuit Design

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Notably, the H-DES algorithm keeps its quantum circuits shallow (few layers of rotations and CNOTs), consistent with this mitigation strategy. This is feasible because the problem structure allows a *low-depth ansatz*, which helps avoid the random unitary regime that causes barren plateaus.

- Local Cost Functions

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H-DES explicitly suggests designing the cost function to be less prone to barren plateaus. The loss is constructed from *local error terms* of the PDE, aligning with the idea of using local (or structured) cost functions to maintain large gradients.

- Restructuring Ansatz Connectivity: the architecture of the ansatz can be constrained to avoid full entanglement at large depth (techniques like residual connections from classical neural networks, *ResNets*)

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H-DES uses a hardware-efficient circuit tailored to the problem (only R_y rotations to generate real-valued states, reflecting the needs of the PDE solution). This reduced ansatz expressivity (and using linear entangling connectivity) prevents the state from exploring irrelevant random subspaces, which is in line with mitigating barren plateaus by constraining the Ansatz.

2. Expressive Ansatz Issues

In Ansätze with extremely high expressivity (often due to many parameters and layers) can represent a very large portion of the unitary space. While expressivity is generally desirable, *over-parameterized or unstructured Ansätze* can produce states that are essentially random, leading to a cost landscape that is almost flat.

Mitigation techniques:

- Problem-Inspired Ansätze: reduce Expressivity
- Entanglement Limiting and Regularization [6]:
  - Partitioning the qubits into separate registers so that not all qubits participate in the cost function
  - Selective inter-layer interactions – entangling only specific pairs of registers at a time, rather than global all-to-all entanglement
  - Adding an entanglement penalty term to the loss (e.g. a term like $\lambda \sum_{\text{cut}} S(\rho_{\text{A}})$ that penalizes entanglement entropy S across certain bipartitions)

- Meta-learning low-entanglement initial states

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In H-DES it is mentioned that Initializing the variational circuit using knowledge of the differential equation (rather than random parameters) could start the optimization closer to the solution, thereby avoiding barren plateaus from the outset.

3. Noise-Induced Barren Plateaus

Gate noise and decoherence can induce barren plateaus, even if the Ansatz and initialization are well-chosen. Noise channels tend to drive the quantum state towards a maximally mixed state as the circuit depth increases. Consider a noise channel $\mathcal{N}$ that acts on qubits (e.g. depolarizing or Pauli error) with some error probability p per gate. After many noisy gates, the state $\rho(\boldsymbol{\theta})$ will approximate the identity matrix $I/2^n$ (for depolarizing noise) or at least lose its dependence on $\boldsymbol{\theta}$. The maximally mixed state is invariant under gates and thus not sensitive to parameter changes.

Mitigation techniques:
• Frequent calibration of quantum hardware can help reduce SPAM errors.
• Characterizing error rates enables compensation techniques.
• *Zero-noise extrapolation* (ZNE) can reduce the impact of state preparation errors.
ZNE follows these three key steps:
i) *Amplify the Noise*: instead of directly reducing noise (which is hard), we systematically increase the noise in a controlled manner. This is done by artificially stretching gate operations or increasing error rates.
ii) *Measure the Results*: the quantum circuit is executed multiple times at different noise levels, and results are collected.
iii) *Extrapolate to Zero Noise*: using classical post-processing techniques (e.g., polynomial fitting), we estimate what the result would be if the noise were zero by extrapolating from the noisy data. Increasing the number of shots (measurements) averages out some of the noise.

Info

ZNE in the context of H-DES? Algorithmic and/or hardware approaches?

References

DLA talk: https://www.youtube.com/watch?v=xQ4ajDc4V3Y

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